# Probleme examen AA

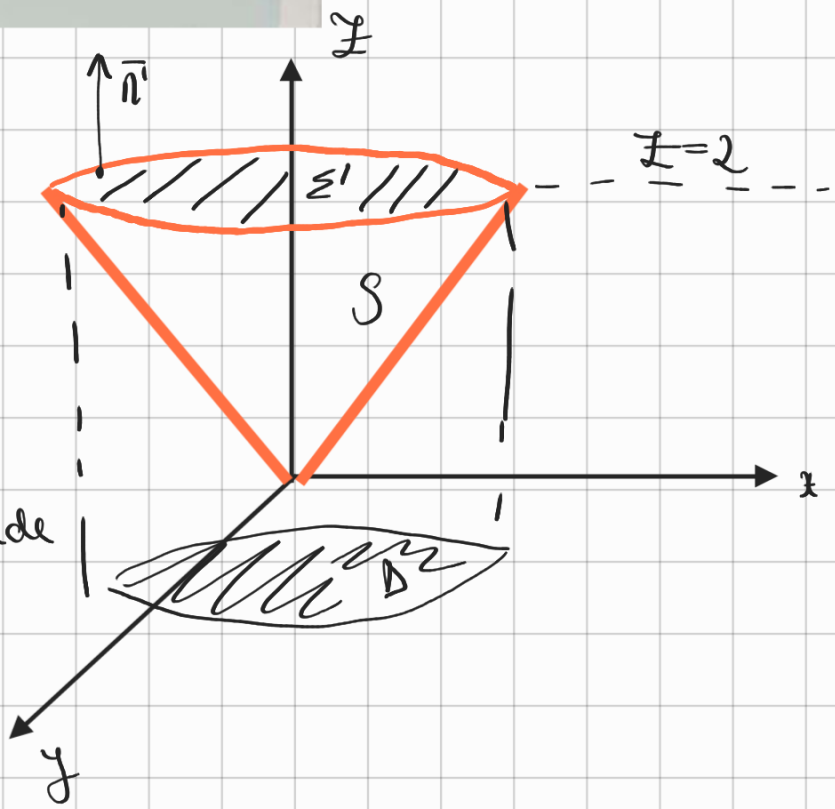


## problema cu flux

Fie câmpul vectorial $\bar{V} = (\overbrace{2x^2y + \sqrt{z^3}}^{P}, \overbrace{\sqrt{z^3} - 2xy^2}^{Q}, \overbrace{z}^{R})$ și suprafața $S = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 = 2x^2 + 2y^2, z \in [0, 2]\}$. Să se determine fluxul câmpului vectorial $\bar{V}$ prin suprafața $S$, orientată după normala ce formează unghi obtuz cu semiaxa pozitivă $Oz$.

$\Sigma_1 = S \cup \Sigma'$

$S$ suprafața conului $z^2 = 2x^2 + 2y^2$, $z \in [0,2]$




$$\Phi_{\Sigma_1} = \int\int_{\Sigma_1} \bar{V} \cdot \bar{n}\, d\sigma \overset{G-O}{=}$$

$$\iiint_{\Omega} \operatorname{div}\bar{V}\, dx\,dy\,dz \text{, unde}$$

$\Omega = \{(x,y,z) \mid 2x^2+2y^2 \le z^2, z \in [0,2]\}$

$$\operatorname{div}\bar{V} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 4xy - 4xy + 1 = 1 \Rightarrow$$

$$\Phi_{\Sigma_1} = \iiint_{\Omega} dx\,dy\,dz$$

$$\Omega: \begin{cases} z \in [\sqrt{2x^2+2y^2}, 2] \\ 2x^2+2y^2 \le 4 \end{cases} \Rightarrow \begin{cases} z \in [\sqrt{2x^2+2y^2}, 2], z = z \\ x^2+y^2 \le 2 \quad \rho \in [0, \sqrt{2}] \\ x = \rho\cos t \quad t \in [0, 2\pi] \\ y = \rho \sin t \quad \text{Jacobian } \rho \end{cases}$$

$$\Rightarrow \Phi_{\Sigma_1} = \iint_D dx\,dy \int_{\sqrt{2x^2+2y^2}}^{2} dz = \iint_D (2 - \sqrt{2x^2+2y^2})\, dx\,dy =$$

$$\int_0^{2\pi} dt \int_0^{\sqrt{2}} (2 - \sqrt{2}\cdot\rho)\cdot\rho\, d\rho = \int_0^{2\pi} \left(\rho^2 - \frac{\sqrt{2}\rho^3}{3}\right)\Big|_0^{\sqrt{2}} dt = \int_0^{2\pi} \left(2 - \frac{4}{3}\right) dt =$$

$$= \int_0^{2\pi} \frac{2}{3}\, dt = \frac{2}{3} t \Big|_0^{4\pi} = \frac{4\pi}{3}$$

$$\Phi_S = \Phi_{\Sigma_1} - \Phi_{\Sigma_1'}$$

$$\Phi_{\Sigma_1'} = \int_{\Sigma_1'} \bar{V}\bar{n}'\, d\sigma = \int_{\Sigma_1'} z\, d\sigma = 2 \iint_{\Delta} dx\, dy = 2 \int_0^{2\pi} dt \int_0^{\sqrt{2}} \rho\, d\rho =$$

$$= 2 \int_0^{2\pi} \frac{\rho^2}{2}\Big|_0^{\sqrt{2}} dt = 2 \int_0^{2\pi} dt = 4\pi \Rightarrow$$

$$\Phi_S = \Phi_{\Sigma_1} - \Phi_{\Sigma_1'} = \frac{4\pi}{3} - 4\pi = \frac{4\pi - 12\pi}{3} = \frac{-8\pi}{3}$$

Fie câmpul vectorial $\bar{V} = \left(\overbrace{\sqrt{z^3 + x^2y^2}}^{P}, \overbrace{\sqrt{z^3 - \frac{2xy^3}{3}}}^{Q}, \overbrace{z}^{R}\right)$ și suprafața $S = \{(x, y, z) \in \mathbb{R}^3 \mid 9z^2 = x^2 + y^2, z \in [0, 1]\}$. Să se determine fluxul câmpului vectorial $\bar{V}$ prin suprafața $S$, orientată după normala ce formează unghi obtuz cu semiaxa pozitivă $Oz$.

$\Sigma_1 = S + \Sigma_1'$

$S$ suprafața conului $9z^2 = x^2 + y^2$, $z \in [0,1]$

$$\Phi_{\Sigma_1} = \int_{\Sigma} \bar{n}\bar{V}\, d\sigma \overset{G-O}{=}$$

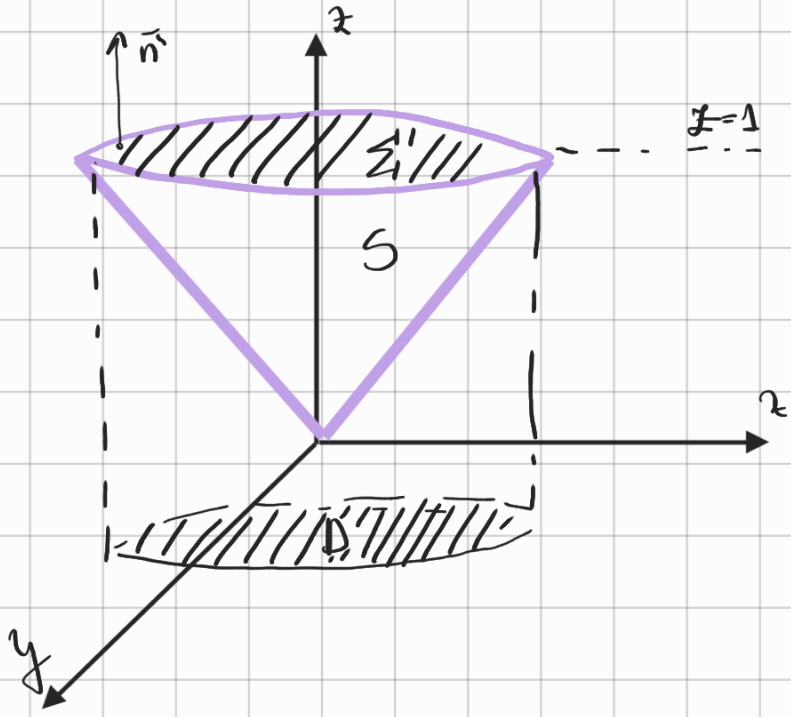
$$= \iiint_{\Omega} \operatorname{div}\bar{V}\, dx\, dy\, dz,$$

unde $\Omega = \{(x,y,z) \mid 9z^2 \geq x^2 + y^2,\ z \in [0,1]\}$

și $\operatorname{div}\bar{V} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} =$

$$= \frac{2xy^2}{2\sqrt{z^3 + x^2y^2}} - \frac{2 \cdot 3xy^2}{3\sqrt{z^3 - \frac{2xy^3}{3}}} + 1 = 1 \Rightarrow$$

$$\Phi_{\Sigma_1} = \iiint_{\Omega} dx\, dy\, dz$$




Jacobian $\rho$

$$\Omega: \begin{cases} z \in \left[\frac{\sqrt{x^2+y^2}}{3}, 1\right], z = z \\ x = \rho\cos t \\ y = \rho\sin t \end{cases} \quad \begin{matrix} t \in [0, 2\pi] \\ \rho^2 \leq 9 \Rightarrow \\ \rho \in [0,3] \end{matrix}$$

$$\Rightarrow \Phi_{\Sigma_1} = \int_0^{2\pi} dt \int_0^3 \left(2 - \frac{\rho}{3}\right) \cdot \rho \, d\rho = \int_0^{2\pi} \left(\frac{\rho^2}{2} - \frac{\rho^3}{9}\Big|_0^3\right) dt = 2\pi\left(\frac{9}{2} - 3\right) =$$

$$= \pi(9-6) = 3\pi$$

$$\boxed{\Phi_S = \Phi_{\Sigma_1} - \Phi_{\Sigma_1'}}$$

$$\Phi_{\Sigma_1'} = \int_{\vec{\Sigma}_1'} \vec{v} \cdot \vec{n}' \, d\sigma = \int_{\Sigma_1'} z \, d\sigma = \iint_D dx\,dy = \text{aria}(D) = 9\pi$$

$$\Rightarrow \quad \Phi_S = 3\pi - 9\pi = -6\pi$$